DP Biology – D4.1 (HL extension)

# Selection and gene pools

- What can change allele frequencies in a population?
- What part does reproduction play in natural selection?

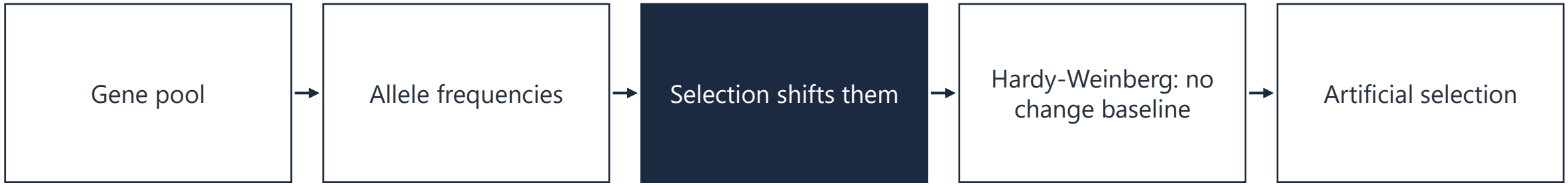
HL statements D4.1.9–15. Textbook pp. 790–797.
Image: a male guppy. 5snake5, CC0.



# Objectives for each class

| Class | Statements | By the end of the class, you can... |
|---|---|---|
| 1 | D4.1.9–10 | Describe gene pools, calculate allele frequencies, and compare separated populations |
| 2 | D4.1.11–12 | Explain how selection changes allele frequencies, and compare three patterns of selection |
| 3 | D4.1.13–15 | Use the Hardy-Weinberg equation and its conditions, and compare artificial with natural selection |

# The big picture: counting alleles



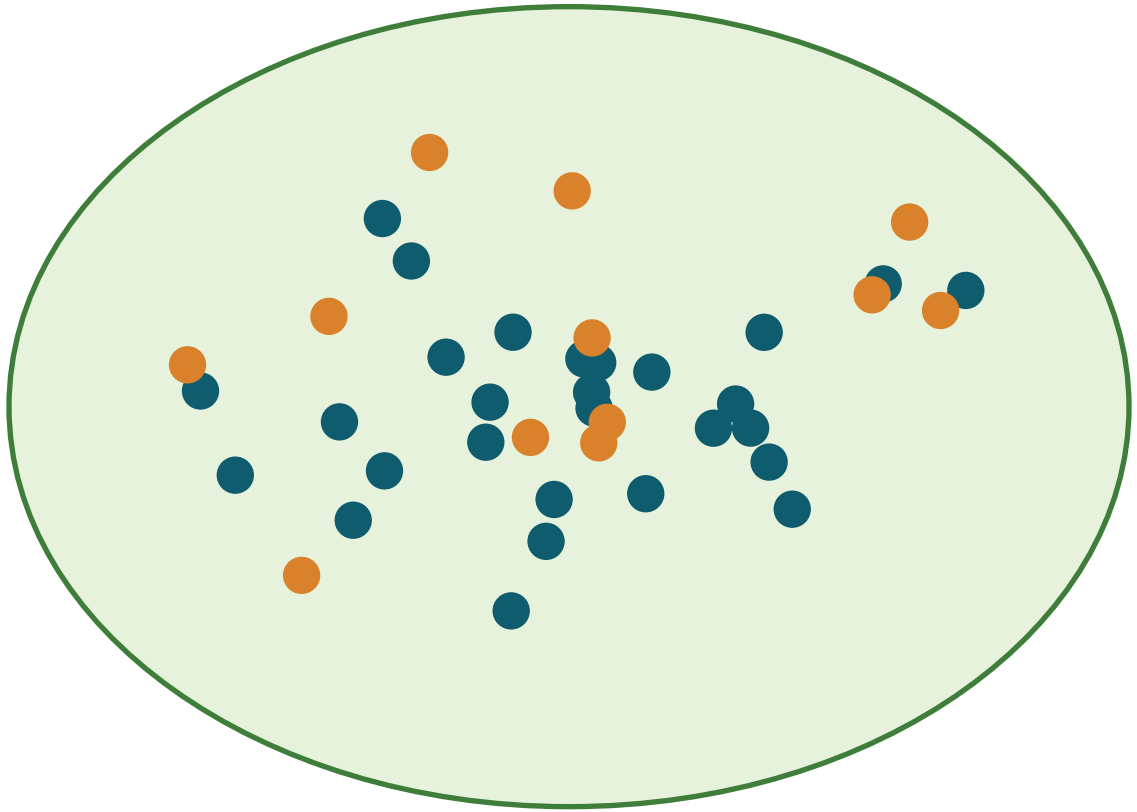

**Evolution** is a change in allele frequencies. By counting alleles, we can detect and measure it.

**Big question**

If only 1 in 20 000 people has albinism, how many carry the allele?

# The gene pool

D4.1.9



gene pool: every allele copy in the population

A **gene pool** is all the genes and alleles in an interbreeding population.

**Allele frequency** = copies of one allele / all copies of the gene.

**Here**

teal 28/40 = 0.7; orange 0.3

# Calculating allele frequency



| Genotype | Number | R copies |
|---|---|---|
| **RR** | 49 | 98 |
| **Rr** | 42 | 42 |
| **rr** | 9 | 0 |
| **Total** | 100 | 140 of 200 |

**So**

Frequency of R = 140 / 200 = 0.7. Frequency of r = 0.3. p + q = 1.

**Key idea**

Each population has its own gene pool, because mating happens mainly within it.

# Allele frequencies differ between populations

D4.1.10

| Allele (approximate frequency) | Northern Europe | East Asia | West Africa |
|---|---|---|---|
| **Lactase persistence** | 0.75 | Near 0 | Near 0 |
| **ALDH2 variant (alcohol flush)** | Near 0 | 0.25 | Near 0 |
| **Sickle cell (HbS)** | Near 0 | Near 0 | Up to 0.15 |

Rounded values from published surveys; frequencies vary within each region.

**Why?**

Separated populations have different mutations, selection pressures and chance events. Malaria selects for HbS carriers where it is common.

# Practice: allele frequencies

D4.1.9–10

**Calculate**

36 AA, 48 Aa, 16 aa: frequency of a?

**Explain**

Why lactase persistence differs by region.

**Define**

Gene pool.

**Use**

allele frequency, gene pool, population, selection pressure

# Working with data: two islands

D4.1.10

| Genotype | Island X | Island Y |
|---|---|---|
| **BB** | 64 | 16 |
| **Bb** | 32 | 48 |
| **bb** | 4 | 36 |

1. Calculate q (b) on each island.
2. Suggest two reasons for the difference.
3. What would migration do over time?

# Check-in: class 1

D4.1.9–10

1. Define gene pool.
2. Define allele frequency.
3. Calculate p if q = 0.35.
4. Explain why separated populations differ.
5. Give an example of a selection pressure changing a frequency.

**How to use this**

Answer without your notes, then check against slides 3–8.

# Selection changes allele frequencies

D4.1.11



Peppered moth, pale form. Chiswick Chap, CC BY-SA 3.0.

Individuals with favourable alleles survive and reproduce more, so those alleles **rise in frequency**. Darwin + genetics = **neo-Darwinism**.

Soot darkens trunks → Dark moths hidden → More dark moths breed → **Dark allele rises**

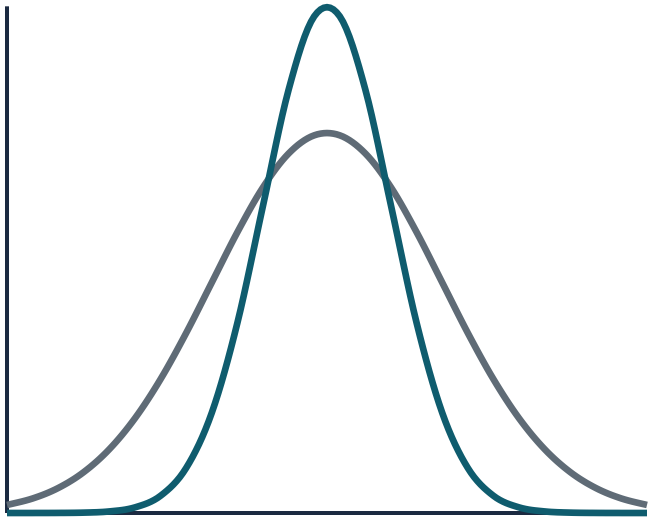
### Reversal

Clean air laws: pale moths hidden again; the dark allele fell.

# Three patterns of selection

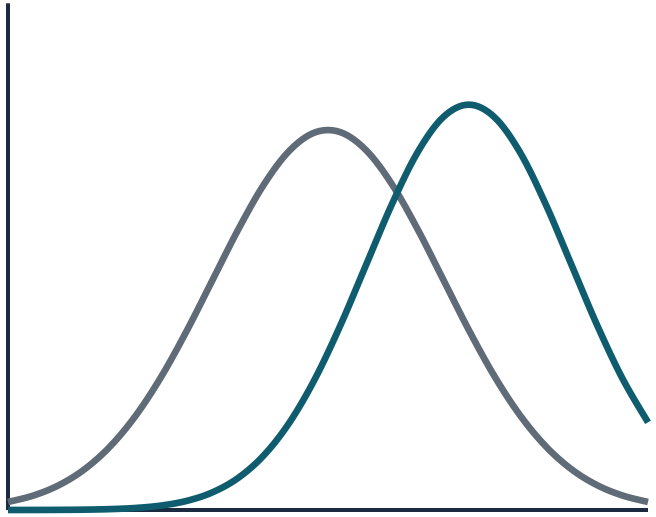
D4.1.12

### Stabilizing



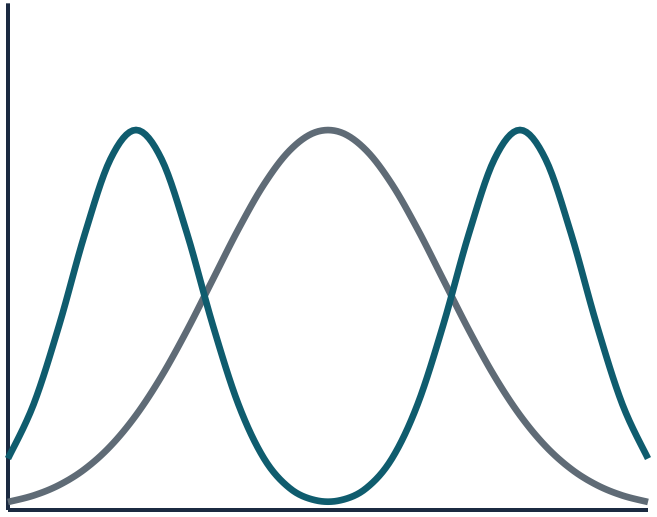
Value of trait

### Directional



Value of trait

### Disruptive



Value of trait

Grey: before selection. Teal: after selection.

| Favours the mean | Favours one extreme | Favours both extremes |
|---|---|---|
| **Human birth weight** | Dark moths; larger turkeys | Finches with very large or small beaks |

# Working with data: birth weight

D4.1.12

Infant survival by birth weight (rounded, classic hospital data pattern).

| Birth weight / kg | Survival / % |
|---|---|
| 1.5 | 60 |
| 2.5 | 92 |
| 3.5 | 98 |
| 4.5 | 93 |
| 5.5 | 80 |

1. Which weight survives best?
2. Name the selection pattern.
3. Suggest why modern medicine weakens it.

# Practice: patterns of selection

D4.1.11–12

**Classify**

Antibiotic resistance in bacteria.

**Classify**

Birds with mid-length wings survive storms best.

**Classify**

Two beak sizes survive; middle ones starve.

### Sketch

Draw before and after curves for each.

# Check-in: class 2

D4.1.11–12

1. Explain how selection changes allele frequencies.
2. Define neo-Darwinism.
3. Describe stabilizing selection with an example.
4. Describe directional selection with an example.
5. Describe disruptive selection with an example.

**How to use this**

Answer without your notes, then check against slides 10–13.

# The Hardy-Weinberg equation

D4.1.13

$$p + q = 1$$

$$p^2 + 2pq + q^2 = 1$$

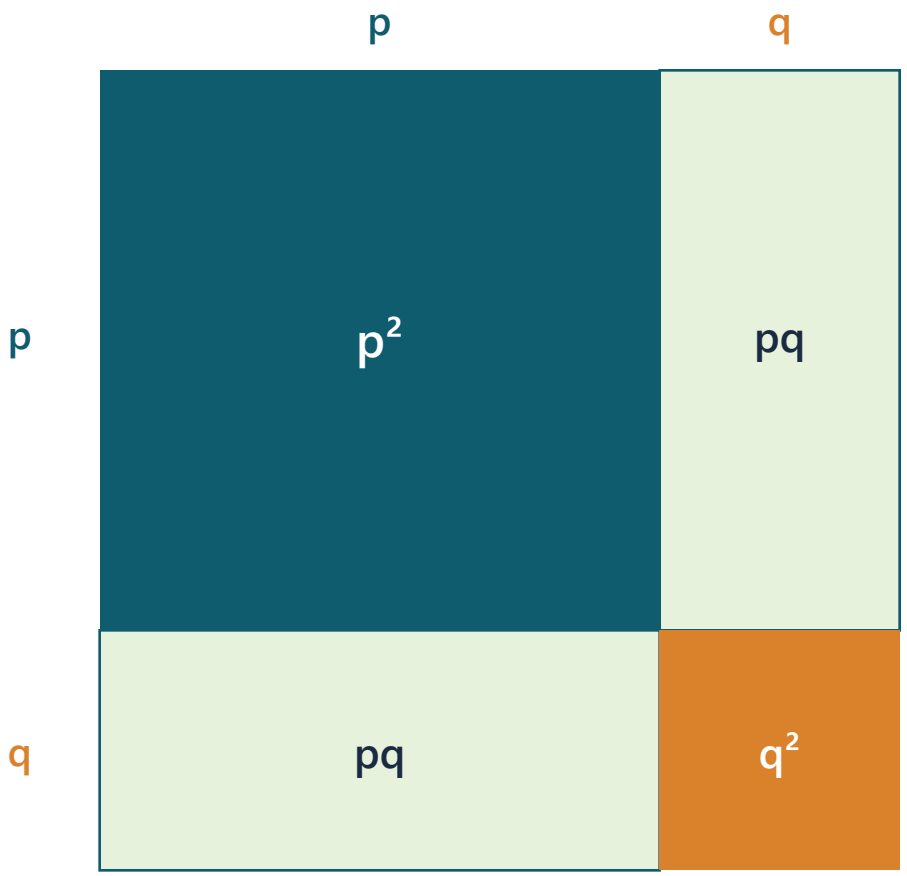
| Symbol | Meaning | Who |
|---|---|---|
| **p, q** | Allele frequencies | Dominant, recessive |
| **$p^2$** | Genotype frequency | Homozygous dominant |
| **2pq** | Genotype frequency | Heterozygous |
| **$q^2$** | Genotype frequency | Homozygous recessive |

**Tip**

Start from the homozygous recessive group: the only genotype visible from the phenotype.

# Why $p^2 + 2pq + q^2$

D4.1.13

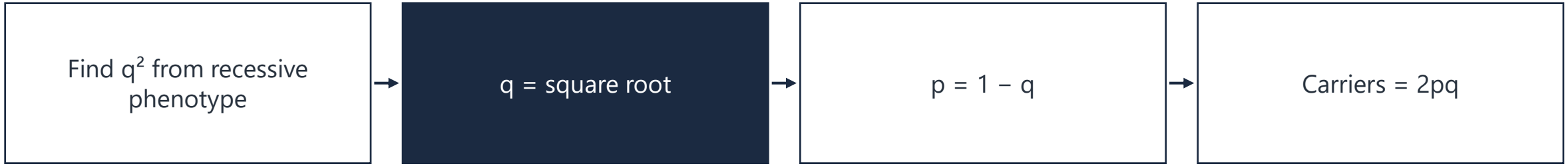



## Big question

Albinism: $q^2$ = 1/20 000, q = 0.007, p = 0.993. Carriers 2pq = 0.014: about 1 in 70 people.

# Worked example step by step

D4.1.13




## Try it

Tay-Sachs (tt) has a frequency of 0.0003 in one population. Calculate q and the frequency of carriers.

# Conditions for Hardy-Weinberg equilibrium

D4.1.14

| Condition | What breaks it |
| --- | --- |
| **Random mating** | Choosing mates by phenotype |
| **No selection** | Some genotypes do better |
| **No mutation** | New alleles arise |
| **No migration** | Individuals move in or out |
| **Very large population** | Genetic drift in small populations |

### Using it

If genotype frequencies do not fit, a condition is broken: the population is evolving.

# Working with data: snails

D4.1.13–14

A population of 1000 snails: 160 yellow (recessive, aa); the rest brown. Invented values.

1. Calculate q.
2. Calculate p.
3. Expected number of heterozygotes?
4. Birds eat more yellow snails. Predict the change in q and name the selection.

# Artificial selection

D4.1.15



Two dogs, one species: selective breeding. Elf, CC BY-SA 3.0.



Bread wheat. Bluemoose, CC BY-SA 3.0.

Dogs from wolves; wheat with large grains.

| | Natural selection | Artificial selection |
|---|---|---|
| **Who selects** | The environment | People, deliberately |
| **Favours** | Survival and reproduction | Traits people want, even harmful |

**Careful**

Antibiotic resistance is natural selection: nobody chose it.

# When artificial selection harms

D4.1.15



A domestic turkey. Irvin calicut, CC BY-SA 3.0.

Turkeys were bred for large breast muscle, so heavy that most cannot mate naturally.

Traits people want can reduce **fitness**.

## Discuss

Is it ethical to breed animals with traits that harm their welfare?

# Practice: HW and artificial selection

D4.1.13–15

**Calculate**

9% are aa: find p, q and 2pq.

**Explain**

Why a small island breaks HW.

**Compare**

Natural and artificial selection.

**Use**

p, q, carrier, drift, equilibrium

# Check-in: class 3

D4.1.13–15

1. State the two Hardy-Weinberg equations.
2. Explain why you start from $q^2$.
3. List the five conditions.
4. Explain what a poor fit means.
5. Distinguish natural and artificial selection.

**How to use this**

Answer without your notes, then check against slides 15–22.

# Common mistakes to avoid

| Not this | But this |
| --- | --- |
| **q = recessive individuals / total.** | $q^2$ is that fraction; q is its root. |
| **Dominant alleles always spread.** | Only selection, drift, mutation or migration change frequencies. |
| **HW describes real populations exactly.** | It is a model; departures show evolution. |
| **Antibiotic resistance is artificial selection.** | No one chose it: natural selection. |

# Summary: selection and gene pools

D4.1.9–15

| | Key idea |
|---|---|
| **D4.1.9** | Gene pool; allele frequency |
| **D4.1.10** | Frequencies differ between populations |
| **D4.1.11** | Selection changes frequencies |
| **D4.1.12** | Stabilizing, directional, disruptive |
| **D4.1.13** | $p + q = 1$; $p^2 + 2pq + q^2 = 1$ |
| **D4.1.14** | Five conditions for equilibrium |
| **D4.1.15** | Artificial selection |

# Check that you can...

1. **Define** gene pool and allele frequency
2. **Explain** how selection changes frequencies
3. **Distinguish** three patterns of selection
4. **Calculate** with Hardy-Weinberg
5. **Outline** equilibrium conditions
6. **Compare** natural and artificial selection

# Exit activity: 5 minutes

**On your own and without notes. Hand this in as you leave.**

1. Define gene pool.

2. In a population, 4% are homozygous recessive. Calculate the frequency of the dominant allele.

3. State why antibiotic resistance is not an example of artificial selection.

# Image credits

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